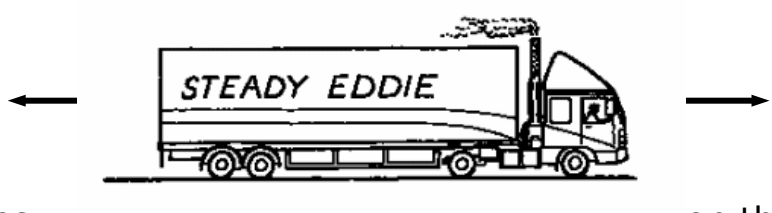
Balanced and Unbalanced Forces

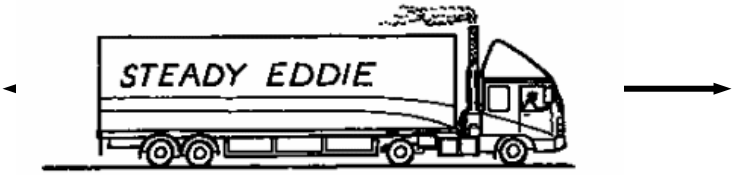
When two forces are balanced:
A stationary object stays stationary.
A moving object moves at the same speed.
An object's shape will stay the same.

When two forces are not balanced:
A moving object will change speed.
A stationary object will move.
An object's shape will change.

Look at the pictures of the truck below. The arrows represent the forces acting on the truck. A longer arrow shows a larger force and a shorter arrow shows a smaller force.



- The forces on the truck are _____.
- The truck will _____.



- The forces acting on the truck are _____.
- The larger force is acting in the _____ direction as the movement of the truck.
- The truck will _____.



- The forces acting on the truck are _____.
- The larger force is acting in the _____ direction to the movement of the truck.
- The truck will _____.

1. How can you tell if the forces acting on an object are balanced?

2. Give two examples of situations where the forces on an object are balanced.

3. How can you tell if the forces acting on an object are unbalanced?

4. Give two examples of situations where the forces on an object are unbalanced.

5. As you sit on your chair, are the forces acting on you balanced or unbalanced? Explain your choice.

6. A bike is slowing down to a stop. Explain why the forces acting on it must be unbalanced.

7. Look at the picture of the two tug of war teams. Below each child is their strength in newtons.
   a. Are the two teams balanced? If not, which team will win?

   b. Try to make the teams balanced.



32N 20N 20N 10N 23N 7N 10N 13N
9N 6N